

# 26020/26021 Chemistry (Polytechnical Foundation) Formula Collection

2026

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# 1 Atoms, Stoichiometry, and Nomenclature

## 1.1 Ctrl+F keywords: isotope, neutron, molar mass, empirical formula, limiting reactant, yield, navngivning, stofmaengde

Use for exam tasks asking for isotope particles, formula mass, mass percent, molecular formula, reaction amounts, limiting reagent, theoretical yield, and ionic compound names.

## 1.2 Core formulas and definitions

### 1.2.1 Isotope notation, mass number, atomic number, neutron count

$${}_Z^AX, \quad A = Z + N, \quad N = A - Z$$

$Z$  is protons,  $N$  is neutrons, and a neutral atom has  $Z$  electrons. Isotopes have the same  $Z$  but different  $N$ .

### 1.2.2 Mole, Avogadro constant, molar mass

$$N = nN_A, \quad N_A = 6.022 \times 10^{23} \text{ mol}^{-1}, \quad n = \frac{m}{M}$$

Convert all masses to moles before using reaction coefficients.  $M$  is found by summing atomic molar masses in the formula.

### 1.2.3 Mass percent and composition

$$w_i = \frac{m_i}{m_{\text{total}}} \times 100\%, \quad w_{\text{element}} = \frac{n_i M_i}{M_{\text{compound}}} \times 100\%$$

Use subscripts in the chemical formula as mole counts of atoms per mole compound.

### 1.2.4 Empirical formula and molecular formula

$$n_i = \frac{m_i}{M_i}, \quad \text{empirical ratio} = \frac{n_i}{\min(n_i)}, \quad M_{\text{molecular}} = kM_{\text{empirical}}$$

Round only after checking near-integer ratios such as  $1.5 \rightarrow 3/2$ . The molecular formula is  $k$  times the empirical formula.

### 1.2.5 Balanced equation stoichiometry

$$\nu_A A + \nu_B B \rightarrow \nu_C C, \quad n_C = n_A \frac{\nu_C}{\nu_A}$$

Stoichiometric coefficients are mole ratios, not mass ratios. Balance atoms and charge before calculating.

### 1.2.6 Limiting reactant and theoretical yield

$$\xi_i = \frac{n_i}{\nu_i}, \quad \xi_{\max} = \min(\xi_i), \quad n_{\text{product}} = \xi_{\max} \nu_{\text{product}}$$

The smallest available reaction extent is the limiting reactant. Convert product moles to mass with  $m = nM$ .

### 1.2.7 Percent yield

$$\% \text{yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100\%$$

Use the same units in numerator and denominator. Percent yield above 100% usually signals a wrong limiting reactant or unit conversion.

### 1.3 Exam recipe: stoichiometry, limiting reactant, gas amount, udbytte

**Use when:** the problem gives reactant masses, solution concentrations, or gas data and asks for product mass, amount, excess, or yield.

**Steps:**

1. Write and balance the reaction.
2. Convert every given reactant to moles.
3. Divide each reactant amount by its coefficient  $\nu_i$ .
4. The smallest  $n_i/\nu_i$  is limiting.
5. Multiply  $\xi_{\max}$  by product coefficient and convert to requested units.

**Fast checks:** atoms and charge balance; answer cannot exceed theoretical yield; units must cancel.

**Common traps:** using grams as mole ratios; forgetting coefficients for  $\text{H}_2$ ,  $\text{O}_2$ , or polyatomic ions; treating excess reactant as limiting.

### 1.4 Exam recipe: empirical formula from mass percent, sum formula

**Use when:** percent composition or masses of elements are given.

**Steps:**

1. Assume 100 g sample if percentages are given.
2. Convert each element mass to moles.
3. Divide all mole amounts by the smallest.
4. Multiply all ratios by 2, 3, ... only if needed to get integers.
5. Use molar mass to scale empirical to molecular formula if requested.

**Fast checks:** empirical formula mass must divide the molecular molar mass approximately by an integer.

### 1.5 Ionic formulas and names

#### 1.5.1 Charge neutrality for ionic compounds

$$\sum_i q_i n_i = 0$$

Subscripts are chosen so total positive and negative charge cancel. Use parentheses when a polyatomic ion appears more than once, for example  $(\text{NH}_4)_2\text{HPO}_4$ .

#### 1.5.2 Common ions, Danish/English search names

| ion                       | name                 | ion                 | name               |
|---------------------------|----------------------|---------------------|--------------------|
| $\text{NH}_4^+$           | ammonium             | $\text{OH}^-$       | hydroxide          |
| $\text{NO}_3^-$           | nitrate              | $\text{NO}_2^-$     | nitrite            |
| $\text{SO}_4^{2-}$        | sulfate/sulfat       | $\text{SO}_3^{2-}$  | sulfite/sulfit     |
| $\text{HSO}_4^-$          | hydrogen sulfate     | $\text{CO}_3^{2-}$  | carbonate/carbonat |
| $\text{PO}_4^{3-}$        | phosphate/fosfat     | $\text{HPO}_4^{2-}$ | hydrogen phosphate |
| $\text{H}_2\text{PO}_4^-$ | dihydrogen phosphate | $\text{CN}^-$       | cyanide            |

These names recur in formula writing, buffers, precipitation, and redox questions.

#### 1.5.3 Binary molecular prefixes

|    |       |   |       |   |      |
|----|-------|---|-------|---|------|
| 1  | mono  | 2 | di    | 3 | tri  |
| 4  | tetra | 5 | penta | 6 | hexa |
| 7  | hepta | 8 | octa  | 9 | nona |
| 10 | deca  |   |       |   |      |

The second element receives the ending “-ide”. Omit “mono” on the first element in common naming.

## 2 Aqueous Reactions, Redox Balancing, and Gases

### 2.1 Ctrl+F keywords: molarity, dilution, net ionic, precipitation, redox, oxidation number

Keywords continued: gas law, partial pressure, ideal gas, gas stoichiometry, pressure tank.

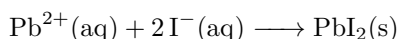
### 2.2 Solutions and aqueous reactions

#### 2.2.1 Molarity, dilution, actual ion concentration

$$c = \frac{n}{V}, \quad n = cV, \quad c_1V_1 = c_2V_2$$

Use liters for  $V$ . For strong electrolytes, multiply by the ion coefficient:  $\text{CaCl}_2 \longrightarrow \text{Ca}^{2+} + 2\text{Cl}^-$  gives  $[\text{Cl}^-] = 2c$ .

#### 2.2.2 Net ionic equation and spectator ions



Remove ions that appear unchanged on both sides. Net ionic equations must conserve atoms and charge.

#### 2.2.3 Precipitation recognition

$$Q_{sp} > K_{sp} \Rightarrow \text{precipitate}, \quad Q_{sp} < K_{sp} \Rightarrow \text{no precipitate yet}$$

Exam wording often says “mixing solutions”, “sparingly soluble”, “bunndfald”, or gives  $K_{sp}$ .

### 2.3 Exam recipe: net ionic precipitation reaction, bunndfald

**Use when:** two aqueous ionic solutions are mixed and a solid product is possible.

**Steps:**

1. Dissociate strong aqueous electrolytes into ions.
2. Pair cations and anions to identify possible insoluble products.
3. Write the solid with phase ( $s$ ).
4. Cancel spectator ions.
5. Check atom and charge balance.

**Common traps:** leaving nitrate or alkali metal spectator ions in the net equation; forgetting ion charges when writing formulas.

### 2.4 Redox balancing

#### 2.4.1 Oxidation and reduction half-reactions

oxidation: electrons produced,      reduction: electrons consumed

Oxidation state increases during oxidation and decreases during reduction. Electron counts must cancel in the total reaction.

#### 2.4.2 Half-reaction balancing in acid or base

balance atoms except H, O,

balance O with  $\text{H}_2\text{O}$ ,

balance H with  $\text{H}^+$ ,

balance charge with  $\text{e}^-$ ,

for base: add  $\text{OH}^-$  to neutralize  $\text{H}^+$ .

The final equation must have no free electrons and must conserve total charge.

## 2.5 Exam recipe: balance redox equation in acidic or basic solution

**Use when:** the problem asks to balance  $\text{MnO}_4^-$ , oxalate, iodide, sulfite, chromium, or other redox reactions.

**Steps:**

1. Split into oxidation and reduction half-reactions.
2. Balance non-H, non-O atoms.
3. Balance O with  $\text{H}_2\text{O}$  and H with  $\text{H}^+$ .
4. Balance charge with electrons.
5. Multiply half-reactions so electrons cancel and add.
6. In basic solution, add  $\text{OH}^-$  to both sides for each  $\text{H}^+$ , form  $\text{H}_2\text{O}$ , then cancel water.

**Fast checks:** same total charge and same atom count on both sides.

## 2.6 Gases

### 2.6.1 Ideal gas law and combined gas law

$$pV = nRT, \quad \frac{p_1 V_1}{T_1} = \frac{p_2 V_2}{T_2} \quad (n \text{ constant})$$

Temperatures must be in kelvin. Use a gas constant matching pressure units.

### 2.6.2 Partial pressure and mole fraction

$$p_i = x_i p_{\text{tot}}, \quad x_i = \frac{n_i}{n_{\text{tot}}}, \quad p_{\text{tot}} = \sum_i p_i$$

Dalton's law applies to ideal gas mixtures. Mole fraction equals pressure fraction.

### 2.6.3 Gas density and molar mass

$$\rho = \frac{m}{V} = \frac{pM}{RT}, \quad M = \frac{\rho RT}{p}$$

This is the fastest route when density, pressure, and temperature of a gas are given.

## 2.7 Exam recipe: gas stoichiometry, gas volume, pressure tank

**Use when:** gas volume, pressure, temperature, or a heated tank appears in a stoichiometry or state-change problem.

**Steps:**

1. Convert temperature to kelvin and volume to liters or cubic meters consistently.
2. Use  $pV = nRT$  to find moles, or the combined gas law if  $n$  is fixed.
3. For reactions, use balanced coefficients after converting gas data to moles.
4. Convert final moles to requested mass, volume, pressure, or yield.

**Common traps:** Celsius in gas laws; using total pressure where partial pressure is required; forgetting gas stoichiometric coefficients.

## 3 Thermodynamics and Thermochemistry

### 3.1 Ctrl+F keywords: enthalpy, Hess, formation enthalpy, combustion, entropy, Gibbs, spontaneity, ligevaegt

Use for heat, work,  $\Delta H$ ,  $\Delta S$ ,  $\Delta G$ , equilibrium constants from thermodynamics, and temperature-dependent spontaneity.

## 3.2 Core formulas

### 3.2.1 First law, heat, and work

$$\Delta U = q + w, \quad w = -p_{\text{ext}} \Delta V$$

Heat into the system is positive. Expansion work is negative for the system when  $\Delta V > 0$ .

### 3.2.2 Constant-pressure heat and enthalpy

$$H = U + pV, \quad q_p = \Delta H$$

At constant pressure, measured heat equals the enthalpy change.  $\Delta H < 0$  is exothermic and  $\Delta H > 0$  is endothermic.

### 3.2.3 Reaction enthalpy from standard formation enthalpies

$$\Delta H_{\text{rxn}}^{\circ} = \sum_{\text{prod}} \nu_i \Delta H_{\text{f},i}^{\circ} - \sum_{\text{react}} \nu_i \Delta H_{\text{f},i}^{\circ}$$

Use coefficients from the balanced equation. Elements in their standard state have  $\Delta H_{\text{f}}^{\circ} = 0$ .

### 3.2.4 Hess's law and thermochemical scaling

$$\Delta H_{\text{overall}}^{\circ} = \sum_k \Delta H_k^{\circ}, \quad 2 \text{ A} \longrightarrow 2 \text{ B} : \Delta H = 2 \Delta H_{\text{A} \longrightarrow \text{B}}$$

Reversing a reaction changes the sign. Multiplying a reaction multiplies  $\Delta H$ ,  $\Delta S$ , and  $\Delta G$ .

### 3.2.5 Combustion enthalpy

$$\Delta H_{\text{comb}}^{\circ} = \sum \nu \Delta H_{\text{f}}^{\circ}(\text{CO}_2, \text{H}_2\text{O}, \dots) - \sum \nu \Delta H_{\text{f}}^{\circ}(\text{fuel}, \text{O}_2)$$

Balance complete combustion to  $\text{CO}_2$  and  $\text{H}_2\text{O}$  first.  $\Delta H_{\text{f}}^{\circ}(\text{O}_2(\text{g})) = 0$ .

### 3.2.6 Entropy change from standard molar entropies

$$\Delta S_{\text{rxn}}^{\circ} = \sum_{\text{prod}} \nu_i S_i^{\circ} - \sum_{\text{react}} \nu_i S_i^{\circ}$$

Use  $\text{J mol}^{-1} \text{K}^{-1}$  for entropy and convert to kJ if combining with  $\Delta H$ .

### 3.2.7 Gibbs free energy and spontaneity

$$\Delta G = \Delta H - T \Delta S$$

$\Delta G < 0$  is spontaneous,  $\Delta G = 0$  is equilibrium, and  $\Delta G > 0$  is nonspontaneous under the stated conditions.

### 3.2.8 Standard free energy and equilibrium constant

$$\Delta G^{\circ} = -RT \ln K, \quad K = e^{-\Delta G^{\circ}/RT}$$

$K > 1$  corresponds to  $\Delta G^{\circ} < 0$ . Use kelvin and consistent energy units.

### 3.2.9 Nonstandard Gibbs free energy

$$\Delta G = \Delta G^{\circ} + RT \ln Q$$

The reaction direction under current conditions is determined by  $\Delta G$ , not by  $\Delta G^{\circ}$  alone.

### 3.3 Exam recipe: calculate $\Delta H^\circ$ , reaction enthalpy, Hess

**Use when:** tables of formation enthalpies, combustion data, or several thermochemical equations are given.

**Steps:**

1. Balance the target reaction exactly as written.
2. For formation data, compute products minus reactants with coefficients.
3. For Hess data, reverse and multiply given reactions to match the target.
4. Add all adjusted enthalpies.

**Fast checks:** elements in standard state contribute zero; exothermic combustion should usually be negative.

### 3.4 Exam recipe: sign of spontaneity from $\Delta H$ , $\Delta S$ , temperature

**Use when:** the question asks if a process is spontaneous at low/high temperature.

| $\Delta H$ | $\Delta S$ | spontaneity |
|------------|------------|-------------|
| –          | +          | all $T$     |
| +          | –          | no $T$      |
| –          | –          | low $T$     |
| +          | +          | high $T$    |

The switch temperature is  $T = \Delta H / \Delta S$  when signs allow a change. Convert entropy units before division.

### 3.5 Common traps and one-line checks

#### 3.5.1 Unit consistency in $\Delta G = \Delta H - T\Delta S$

$$\text{kJ mol}^{-1} - \text{K} \cdot \text{kJ mol}^{-1} \text{K}^{-1}$$

Do not mix  $\Delta H$  in kJ with  $\Delta S$  in J. Convert one of them first.

## 4 Electronic Structure and Periodic Table

### 4.1 Ctrl+F keywords: quantum number, electron configuration, orbital, ionization energy, atomic radius, ionic radius, electronegativity

Use for questions about electron configurations, allowed quantum numbers, periodic trends, isoelectronic ions, and light-energy formulas.

### 4.2 Light and quantization

#### 4.2.1 Photon energy, wavelength, frequency

$$E = h\nu = \frac{hc}{\lambda}$$

Shorter wavelength means higher photon energy. Use SI units unless the data table specifies otherwise.

#### 4.2.2 Bohr energy levels and transitions

$$E_n = -\frac{R_H}{n^2}, \quad \Delta E = E_{n_f} - E_{n_i}$$

Emission releases energy when the electron moves to lower  $n$ . Absorption requires positive energy input.

#### 4.2.3 De Broglie wavelength

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

Use for matter-wave questions involving particle mass and speed.

## 4.3 Quantum numbers and orbitals

### 4.3.1 Allowed quantum numbers

| symbol   | allowed values        | meaning  |
|----------|-----------------------|----------|
| $n$      | $1, 2, 3, \dots$      | shell    |
| $\ell$   | $0, 1, \dots, n-1$    | subshell |
| $m_\ell$ | $-\ell, \dots, +\ell$ | orbital  |
| $m_s$    | $\pm \frac{1}{2}$     | spin     |

Invalid sets usually break one of these allowed-value rules.

### 4.3.2 Subshell labels and capacities

$$\ell = 0, 1, 2, 3 \leftrightarrow s, p, d, f, \quad \# \text{orbitals} = 2\ell + 1, \quad \#e^- = 2(2\ell + 1)$$

Thus  $s, p, d, f$  hold 2, 6, 10, 14 electrons.

### 4.3.3 Electron configuration filling order

$$1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p, 5s, 4d, \dots$$

Fill lower energy orbitals first, obey Pauli exclusion, and maximize unpaired spins within degenerate orbitals.

## 4.4 Exam recipe: write electron configuration or identify ion configuration

**Use when:** the problem asks for electron configuration, valence electrons, or which ion/atom has a given configuration.

**Steps:**

1. Count electrons: neutral atom =  $Z$ , ion =  $Z - \text{charge}$ .
2. Fill orbitals in Aufbau order.
3. For cations of transition metals, remove  $ns$  electrons before  $(n-1)d$ .
4. Check total electron count.

**Common traps:** removing  $3d$  before  $4s$  for transition-metal cations; forgetting ion charge sign.

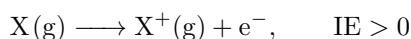
## 4.5 Periodic trends

### 4.5.1 Atomic radius trend

across period: decreases,      down group: increases

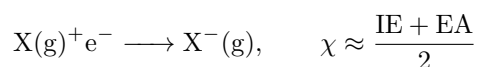
Increasing effective nuclear charge contracts atoms across a period. New shells dominate down a group.

### 4.5.2 Ionization energy trend



Ionization energy generally increases across a period and decreases down a group. Filled and half-filled subshells can create exceptions.

### 4.5.3 Electron affinity and electronegativity



Electronegativity increases toward fluorine. Use it to judge bond polarity and likely formal-charge placement.

### 4.5.4 Isoelectronic ionic radius

same electron count: larger  $Z \Rightarrow$  smaller radius

For  $\text{N}^{3-}$ ,  $\text{O}^{2-}$ ,  $\text{F}^-$ ,  $\text{Na}^+$ ,  $\text{Mg}^{2+}$ , the highest nuclear charge is smallest.

## 4.6 Exam recipe: rank atomic or ionic radius, ionization energy

**Use when:** the question asks for largest/smallest radius or highest/lowest ionization energy.

**Steps:**

1. Check whether species are isoelectronic.
2. If isoelectronic, rank radius by nuclear charge: larger  $Z$  is smaller.
3. If neutral atoms, use period/group trends.
4. For ionization energy, highest is usually up and right in the periodic table.

**Fast checks:** cations are smaller than their neutral atoms; anions are larger than their neutral atoms.

## 5 Bonding, Lewis Structures, Formal Charge, and VSEPR

### 5.1 Ctrl+F keywords: Lewis, formal charge, resonance, VSEPR, bond angle, planar, polarity, dipole, hybridization

Use for drawing or checking Lewis structures, choosing resonance forms, molecular geometry, bond angles, polarity, and planar/non-planar questions.

### 5.2 Lewis structures

#### 5.2.1 Valence electron count

$$N_{\text{val}} = \sum_i N_{\text{val},i} - q$$

For an ion, subtract positive charge and add negative charge. Total Lewis electrons must equal  $N_{\text{val}}$ .

#### 5.2.2 Formal charge

$$\text{FC} = V - \left( N_{\text{nonbonding}} + \frac{1}{2} N_{\text{bonding}} \right)$$

Preferred structures usually minimize formal charges and place negative formal charge on more electronegative atoms.

#### 5.2.3 Resonance structures

resonance forms differ in electron placement, not atom placement

Equivalent resonance forms indicate delocalization and equalized bond lengths. Do not move nuclei when drawing resonance.

#### 5.2.4 Octet, expanded octet, electron-deficient exceptions

period 2 atoms normally obey octet, period 3+ may expand octet

Hydrogen has a duet. Boron can be electron deficient; sulfur, phosphorus, iodine, and xenon can appear hypervalent in course examples.

### 5.3 Exam recipe: Lewis structure and formal charge

**Use when:** the problem asks for nitrate, oxalate, resonance, formal charge, or a valid Lewis structure.

**Steps:**

1. Count total valence electrons, including ion charge.
2. Choose the least electronegative central atom, never hydrogen.
3. Connect atoms with single bonds and complete outer octets.
4. Put leftover electrons on the central atom if allowed.

5. If the central atom lacks an octet, form multiple bonds.
6. Calculate formal charges and choose the best resonance form.

**Fast checks:** electron count matches; total formal charge equals ion charge; second-period atoms do not exceed octet.

## 5.4 VSEPR geometry

### 5.4.1 Steric number and electron domains

$$\text{SN} = \# \text{bonded atoms} + \# \text{lone pairs on central atom}$$

Multiple bonds count as one electron domain in VSEPR. Lone pairs compress bond angles.

### 5.4.2 Common VSEPR geometries and bond angles

| SN | lone pairs | molecular geometry   | angle     |
|----|------------|----------------------|-----------|
| 2  | 0          | linear               | 180°      |
| 3  | 0          | trigonal planar      | 120°      |
| 4  | 0          | tetrahedral          | 109.5°    |
| 4  | 1          | trigonal pyramidal   | ≈ 107°    |
| 4  | 2          | bent                 | ≈ 105°    |
| 5  | 0          | trigonal bipyramidal | 90°, 120° |
| 6  | 0          | octahedral           | 90°       |
| 6  | 1          | square pyramidal     | ≈ 90°     |
| 6  | 2          | square planar        | 90°       |

Exam wording often asks “planar”, “non-planar”, “bond angle”, “pyramidal”, or “bent”.

### 5.4.3 Planar molecule recognition

linear, trigonal planar, square planar, and many aromatic rings are planar

Tetrahedral and trigonal pyramidal centers are non-planar around that atom. A molecule with all atoms in one plane is planar.

### 5.4.4 Molecular polarity

$$\vec{\mu}_{\text{mol}} = \sum_i \vec{\mu}_i$$

Polar bonds can cancel in symmetric geometries. Lone pairs often prevent cancellation and make the molecule polar.

## 5.5 Exam recipe: molecular geometry, bond angle, polarity

**Use when:** the problem asks for shape, bond angle, planar/non-planar, or dipole.

**Steps:**

1. Draw the Lewis structure and count lone pairs on the central atom.
2. Compute steric number.
3. Select VSEPR geometry from the table.
4. Adjust ideal angle downward if lone pairs are present.
5. For polarity, add bond dipoles by symmetry.

**Common traps:** counting a double bond as two VSEPR domains; deciding polarity from bond polarity alone; ignoring lone pairs.

## 5.6 Bonding formulas

### 5.6.1 Bond polarity from electronegativity

$$\Delta\chi = |\chi_A - \chi_B|$$

Larger electronegativity difference gives a more polar bond. Geometry decides whether the whole molecule is polar.

### 5.6.2 Molecular orbital bond order

$$\text{bond order} = \frac{N_{\text{bonding}} - N_{\text{antibonding}}}{2}$$

Higher bond order usually means shorter, stronger bonds. Zero bond order predicts no stable bond in simple MO treatment.

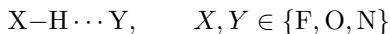
## 6 Intermolecular Forces, Solids, Liquids, and Colligative Properties

### 6.1 Ctrl+F keywords: vapor pressure, hydrogen bonding, boiling point, unit cell, BCC, FCC, density, freezing point depression, Raoult

Use for intermolecular force ranking, vapor pressure, crystal unit-cell density, cubic radius relations, and colligative-property calculations.

### 6.2 Intermolecular forces and phase behavior

#### 6.2.1 Hydrogen bonding recognition



Hydrogen bonding requires H bonded to F, O, or N. It raises boiling points and lowers vapor pressure relative to similar molecules.

#### 6.2.2 Dispersion and dipole forces

all molecules: dispersion,      polar molecules: dipole-dipole

Larger molar mass and surface area increase dispersion forces. Stronger intermolecular forces usually mean higher boiling point and lower vapor pressure.

#### 6.2.3 Like dissolves like

polar dissolves polar,      nonpolar dissolves nonpolar

Solubility is favored when solute-solvent interactions can replace solute-solute and solvent-solvent interactions.

### 6.3 Vapor pressure and phase changes

#### 6.3.1 Clausius-Clapeyron equation

$$\ln P = -\frac{\Delta H_{\text{vap}}}{RT} + C$$

A plot of  $\ln P$  versus  $1/T$  has slope  $-\Delta H_{\text{vap}}/R$ . Vapor pressure increases with temperature.

#### 6.3.2 Two-point vapor pressure equation

$$\ln \left( \frac{P_2}{P_1} \right) = -\frac{\Delta H_{\text{vap}}}{R} \left( \frac{1}{T_2} - \frac{1}{T_1} \right)$$

Use kelvin. This form calculates  $P_2$ ,  $T_2$ , or  $\Delta H_{\text{vap}}$  from two states.

#### 6.3.3 Phase transition enthalpies

$$\Delta H_{\text{sub}} = \Delta H_{\text{fus}} + \Delta H_{\text{vap}}$$

Fusion, vaporization, and sublimation are endothermic in the forward direction.

## 6.4 Exam recipe: vapor pressure, boiling point, intermolecular force ranking

**Use when:** molecules are compared by vapor pressure, boiling point, volatility, or hydrogen bonding.

**Steps:**

1. Identify whether hydrogen bonding is possible.
2. Identify molecular polarity and approximate molar mass.
3. Stronger forces imply higher boiling point and lower vapor pressure.
4. If numerical  $P, T, \Delta H_{\text{vap}}$  data are given, use Clausius-Clapeyron.

**Common traps:** assuming any molecule containing oxygen hydrogen-bonds; the H must be bonded to O, N, or F.

## 6.5 Crystals and unit cells

### 6.5.1 Atoms per cubic unit cell

| cell type                | $Z$ | packing fraction |
|--------------------------|-----|------------------|
| simple cubic             | 1   | 0.52             |
| body-centered cubic, BCC | 2   | 0.68             |
| face-centered cubic, FCC | 4   | 0.74             |

Corner atoms contribute 1/8, face atoms 1/2, and body-center atoms 1.

### 6.5.2 Unit-cell density

$$\rho = \frac{ZM}{N_A a^3}$$

$Z$  is atoms or formula units per cell and  $a$  is cubic edge length. Convert  $a$  to centimeters if  $\rho$  is in  $\text{g cm}^{-3}$ .

### 6.5.3 Cubic radius relations

| cell         | edge-radius relation |
|--------------|----------------------|
| simple cubic | $a = 2r$             |
| BCC          | $\sqrt{3}a = 4r$     |
| FCC          | $\sqrt{2}a = 4r$     |

Use the diagonal that actually touches atoms: edge for simple cubic, body diagonal for BCC, face diagonal for FCC.

## 6.6 Exam recipe: unit cell density, volume, radius

**Use when:** the problem gives BCC/FCC/simple cubic, density, atom radius, molar mass, or unit-cell volume.

**Steps:**

1. Identify  $Z$  from the cell type.
2. Use the radius relation to find  $a$ , or use  $a^3 = V_{\text{cell}}$ .
3. Insert into  $\rho = ZM/(N_A a^3)$ .
4. Convert units carefully.

**Fast checks:**  $1 \text{ pm} = 10^{-10} \text{ cm}$ ;  $1 \text{ nm} = 10^{-7} \text{ cm}$ .

## 6.7 Colligative properties

### 6.7.1 Raoult's law and vapor-pressure lowering

$$P_{\text{solvent}} = X_{\text{solvent}} P_{\text{solvent}}^{\circ}, \quad \Delta P = X_{\text{solute}} P_{\text{solvent}}^{\circ}$$

A nonvolatile solute lowers solvent vapor pressure in proportion to solvent mole fraction.

### 6.7.2 Molality and freezing point depression

$$m = \frac{n_{\text{solute}}}{m_{\text{solvent}}(\text{kg})}, \quad \Delta T_f = iK_f m, \quad \Delta T_b = iK_b m$$

$i$  is the van't Hoff factor, approximately the number of dissolved ions for complete dissociation.

## 6.8 Exam recipe: freezing point depression, boiling point elevation

**Use when:** the problem gives salt mass, solvent mass,  $K_f$ ,  $K_b$ , or asks for freezing/boiling temperature.

**Steps:**

1. Convert solute mass to moles.
2. Convert solvent mass to kilograms and compute molality.
3. Choose  $i$  from dissociation stoichiometry if relevant.
4. Compute  $\Delta T$  and apply the sign: freezing decreases, boiling increases.

**Common traps:** using solution mass instead of solvent mass; forgetting  $i$  for salts such as NaBr.

## 7 Kinetics and Chemical Equilibrium

### 7.1 Ctrl+F keywords: rate law, reaction order, first order, second order, half-life, Arrhenius, catalyst, equilibrium, Le Chatelier, $K_p$

Use for concentration-time data, identifying order, half-life, activation energy, catalyst statements, equilibrium expressions,  $Q$  versus  $K$ , and gas equilibria.

### 7.2 Kinetics formulas

#### 7.2.1 Reaction rate from stoichiometry

$$aA + bB \rightarrow cC + dD$$
$$r = -\frac{1}{a} \frac{d[A]}{dt} = -\frac{1}{b} \frac{d[B]}{dt} = \frac{1}{c} \frac{d[C]}{dt} = \frac{1}{d} \frac{d[D]}{dt}$$

Divide by coefficients so every species reports the same reaction rate.

#### 7.2.2 Rate law and overall order

$$r = k[A]^m[B]^n, \quad \text{overall order} = m + n$$

Orders are found experimentally unless the problem states an elementary step.

#### 7.2.3 Integrated rate laws

| order | linear plot                              | slope |
|-------|------------------------------------------|-------|
| 0     | $[A]_t = -kt + [A]_0$                    | $-k$  |
| 1     | $\ln[A]_t = -kt + \ln[A]_0$              | $-k$  |
| 2     | $\frac{1}{[A]_t} = kt + \frac{1}{[A]_0}$ | $k$   |

Use the plot or transformed data that is linear to identify the order.

#### 7.2.4 Half-life

$$t_{1/2}^{(0)} = \frac{[A]_0}{2k}, \quad t_{1/2}^{(1)} = \frac{\ln 2}{k}, \quad t_{1/2}^{(2)} = \frac{1}{k[A]_0}$$

Only first-order half-life is independent of initial concentration.

### 7.2.5 Arrhenius equation

$$k = Ae^{-E_a/RT}, \quad \ln k = -\frac{E_a}{R} \frac{1}{T} + \ln A$$

A plot of  $\ln k$  versus  $1/T$  has slope  $-E_a/R$ .

### 7.2.6 Two-temperature Arrhenius relation

$$\ln \left( \frac{k_2}{k_1} \right) = -\frac{E_a}{R} \left( \frac{1}{T_2} - \frac{1}{T_1} \right)$$

Use kelvin. If  $T_2 > T_1$ , then usually  $k_2 > k_1$ .

### 7.2.7 Catalyst effect

$$E_{a,cat} < E_{a,uncat}, \quad K_{eq} \text{ unchanged}$$

A catalyst speeds forward and reverse reactions by lowering activation barriers. It does not change  $\Delta G^\circ$  or the equilibrium constant.

## 7.3 Exam recipe: determine reaction order from concentration-time data

**Use when:** a table of concentration versus time is given, or answer options show first/second order.

**Steps:**

1. Test which transformed quantity is linear:  $[A]$ ,  $\ln[A]$ , or  $1/[A]$ .
2. Select zero, first, or second order from the linear plot.
3. Compute  $k$  from the slope with the correct sign.
4. Use the matching integrated law for prediction.

**Fast checks:** first-order equal time intervals give equal fractional decrease; second-order  $1/[A]$  increases linearly.

## 7.4 Equilibrium formulas

### 7.4.1 Reaction quotient and equilibrium constant

$$\alpha A + \beta B \rightleftharpoons \gamma C + \delta D, \quad Q = \frac{a_C^\gamma a_D^\delta}{a_A^\alpha a_B^\beta}$$

At equilibrium,  $Q = K$ . Pure solids, pure liquids, and the solvent are omitted.

### 7.4.2 Direction from $Q$ and $K$

$$Q < K : \text{forward}, \quad Q = K : \text{equilibrium}, \quad Q > K : \text{reverse}$$

The system shifts to make  $Q$  approach  $K$ .

### 7.4.3 Gas equilibrium $K_p$

$$K_p = \frac{(p_C)^\gamma (p_D)^\delta}{(p_A)^\alpha (p_B)^\beta}$$

Use partial pressures with stoichiometric powers. Omit pure condensed phases.

### 7.4.4 Relation between $K_p$ and $K_c$

$$K_p = K_c (RT)^{\Delta n_{\text{gas}}}$$

$\Delta n_{\text{gas}}$  is gaseous product coefficients minus gaseous reactant coefficients.

### 7.4.5 Le Chatelier shift rules

disturbance  $\Rightarrow$  shift that reduces the disturbance

Adding reactant shifts toward products; removing product shifts toward products. Decreasing volume favors the side with fewer gas moles.

## 7.5 Exam recipe: equilibrium expression and Le Chatelier

**Use when:** the problem asks for  $K$ ,  $Q$ ,  $K_p$ , direction of shift, or statement validity.

**Steps:**

1. Write products over reactants with coefficients as exponents.
2. Omit pure solids, pure liquids, and solvent.
3. For gases use partial pressures for  $K_p$ .
4. Compare  $Q$  with  $K$  for direction, or apply the shift rule for disturbances.

**Common traps:** including solids in  $K$ ; reversing  $Q < K$  direction; treating catalysts as changing equilibrium.

## 8 Organic Chemistry, Functional Groups, Chirality, and Polymers

### 8.1 Ctrl+F keywords: functional group, alcohol, ether, ester, amide, carboxylic acid, aromatic, chirality, isomer, polymerization

Use for identifying organic functional groups, aromatic rings, isomers, chiral centers, amino acids, peptides, and polymer repeat units.

### 8.2 Hydrocarbons and unsaturation

#### 8.2.1 Alkane, alkene, alkyne formulas

alkane:  $C_nH_{2n+2}$ , alkene:  $C_nH_{2n}$ , alkyne:  $C_nH_{2n-2}$

Alkenes contain  $C=C$  and alkynes contain  $C\equiv C$ . Rings reduce hydrogen count like one double bond.

#### 8.2.2 Degree of unsaturation, double bond equivalent

$$DBE = \frac{2C + 2 + N - H - X}{2}$$

Oxygen does not affect DBE. Each ring or  $\pi$ -bond counts as one DBE; benzene has four DBE.

#### 8.2.3 Aromatic ring recognition

$C_6H_6$  ring motif, phenyl group:  $C_6H_5-$

A benzene ring is planar and aromatic. Aromatic rings often appear together with alcohol, carboxylic acid, amide, or ester groups.

## 8.3 Functional group lookup

### 8.3.1 Functional group recognition table

| group           | structural motif             |
|-----------------|------------------------------|
| alcohol         | $R-OH$                       |
| ether           | $R-O-R'$                     |
| aldehyde        | $R-CHO$                      |
| ketone          | $R-C(=O)-R'$                 |
| carboxylic acid | $R-COOH$                     |
| ester           | $R-C(=O)-O-R'$               |
| amine           | $R-NH_2, R_2NH, R_3N$        |
| amide           | $R-C(=O)-NH_2, R-C(=O)-NR_2$ |
| nitrile         | $R-C\equiv N$                |

Carbonyl-containing groups are distinguished by what is attached directly to the carbonyl carbon.

### 8.3.2 Ester, ether, amide, amine distinction

ester:  $\text{C}(=\text{O})\text{O}$ , ether:  $\text{C}-\text{O}-\text{C}$ , amide:  $\text{C}(=\text{O})\text{N}$ , amine:  $\text{C}-\text{N}$

Look for the carbonyl first. A nitrogen next to a carbonyl is an amide, not an amine.

## 8.4 Exam recipe: identify functional groups in a molecule

**Use when:** the exam shows an organic structure and asks which functional groups are present.

**Steps:**

1. Mark every heteroatom O, N, halogen, and every carbonyl  $\text{C}=\text{O}$ .
2. Classify carbonyl groups by the atom attached to the carbonyl carbon.
3. Identify aromatic rings separately.
4. Check answer options for extra groups not present.

**Common traps:** confusing ester with ether; confusing amide with amine; missing carboxylic acid because it contains both  $\text{C}=\text{O}$  and  $\text{OH}$ .

## 8.5 Isomerism, chirality, and biomolecules

### 8.5.1 Structural isomers

same molecular formula, different connectivity

Structural isomers can have different names, boiling points, and functional groups.

### 8.5.2 Chiral center

$\text{C}^*$  = tetrahedral carbon attached to four different groups

A molecule with one chiral center is chiral. Symmetry can make molecules with multiple stereocenters achiral.

### 8.5.3 Amino acids and peptide bond

$\text{H}_2\text{N}-\text{CHR}-\text{COOH}$ , peptide bond =  $\text{C}(=\text{O})-\text{NH}$

Amino acids contain amine and carboxylic acid groups. Peptides are connected by amide bonds.

### 8.5.4 Degree of polymerization

$$n = \frac{M_{\text{polymer}}}{M_{\text{repeat}}}$$

The degree of polymerization is the number of repeat units in the chain.

## 8.6 Exam recipe: polymer repeat unit and degree of polymerization

**Use when:** polymer molar mass and repeat-unit mass are given.

**Steps:**

1. Identify the repeat unit.
2. Compute or read  $M_{\text{repeat}}$ .
3. Divide polymer molar mass by repeat-unit molar mass.

**Fast checks:**  $n$  is dimensionless and should be near an integer for idealized exam data.

## 9 Acids, Bases, Buffers, Titration, and Solubility Products

### 9.1 Ctrl+F keywords: pH, pOH, weak acid, $K_a$ , $pK_a$ , buffer, Henderson, phosphate, titration, $K_{sp}$ , common ion, oploeselighed

Use for acid strength, pH calculations, buffers, phosphate systems, titration equivalence, solubility, precipitation, and common-ion tasks.

### 9.2 pH and acid-base constants

#### 9.2.1 Water, pH, and pOH

$$K_w = [\text{H}_3\text{O}^+][\text{OH}^-] = 1.0 \times 10^{-14}, \quad \text{pH} = -\log[\text{H}_3\text{O}^+], \quad \text{pH} + \text{pOH} = 14$$

At 298 K, neutral water has pH 7. A decrease of one pH unit means ten times larger  $[\text{H}_3\text{O}^+]$ .

#### 9.2.2 Strong acid and strong base pH

$$[\text{H}_3\text{O}^+] \approx \nu_{\text{H}^+} c_{\text{acid}}, \quad [\text{OH}^-] \approx \nu_{\text{OH}^-} c_{\text{base}}$$

Strong acids and bases are treated as fully dissociated. Include stoichiometric coefficients, for example  $\text{Ba}(\text{OH})_2$  gives  $2\text{OH}^-$ .

#### 9.2.3 Weak acid and weak base constants

$$K_a = \frac{[\text{A}^-][\text{H}_3\text{O}^+]}{[\text{HA}]}, \quad K_b = \frac{[\text{HB}^+][\text{OH}^-]}{[\text{B}]}$$

Larger  $K_a$  or smaller  $pK_a$  means stronger acid. Larger  $K_b$  or smaller  $pK_b$  means stronger base.

#### 9.2.4 Conjugate acid-base relation

$$K_a K_b = K_w, \quad pK_a + pK_b = 14$$

Use this for conjugate acid/base pairs at 298 K.

#### 9.2.5 Weak acid and weak base approximations

$$[\text{H}_3\text{O}^+] \approx \sqrt{K_a c_0}, \quad [\text{OH}^-] \approx \sqrt{K_b c_0}$$

The approximation is valid when  $x/c_0 \lesssim 5\%$ . Otherwise solve the quadratic.

### 9.3 Exam recipe: pH of weak acid or weak base

**Use when:** a weak acid/base concentration and  $K_a$ ,  $K_b$ ,  $pK_a$ , or  $pK_b$  are given.

**Steps:**

1. Write the equilibrium and initial concentration  $c_0$ .
2. Convert  $pK$  to  $K$  if needed:  $K = 10^{-pK}$ .
3. Use  $x \approx \sqrt{Kc_0}$  for  $[\text{H}_3\text{O}^+]$  or  $[\text{OH}^-]$ .
4. Convert to pH or pOH.
5. Check  $x/c_0$ ; solve exactly if too large.

**Common traps:** reporting pOH as pH for weak bases; forgetting  $pH = 14 - pOH$ .

### 9.4 Buffers and titration

#### 9.4.1 Henderson-Hasselbalch buffer equation

$$\text{pH} = pK_a + \log\left(\frac{[\text{A}^-]}{[\text{HA}]}\right)$$

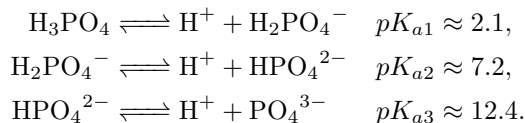
Use mole ratios instead of concentrations when acid and base are in the same final volume.

### 9.4.2 Buffer after adding strong acid or base



Do stoichiometric neutralization first, then apply Henderson-Hasselbalch to the remaining conjugate pair.

### 9.4.3 Phosphoric acid phosphate system



Choose the conjugate pair whose  $pK_a$  is closest to the target or observed pH.

### 9.4.4 Titration equivalence point

$$n_{\text{acid equivalents}} = n_{\text{base equivalents}}$$

At equivalence, stoichiometric acid-base equivalents have reacted. The pH depends on the salt left after neutralization.

## 9.5 Exam recipe: buffer pH, phosphate buffer, Henderson

**Use when:** both weak acid and conjugate base are present, or a buffer receives added strong acid/base.

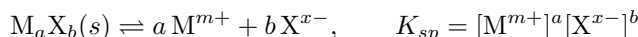
**Steps:**

1. Identify the conjugate pair and the relevant  $pK_a$ .
2. Convert all given concentrations/volumes to moles.
3. If strong acid/base is added, neutralize stoichiometrically first.
4. Insert remaining  $n_{A^-}/n_{HA}$  into Henderson-Hasselbalch.

**Fast checks:** when acid and base amounts are equal,  $pH = pK_a$ ; buffer pH should be near  $pK_a$ .

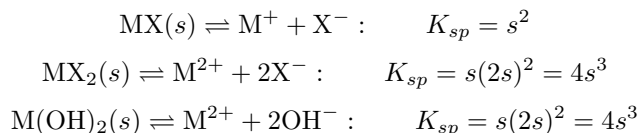
## 9.6 Solubility products

### 9.6.1 Solubility product expression



The pure solid is omitted. Powers come from dissolution stoichiometry.

### 9.6.2 Molar solubility patterns



Convert molar solubility  $s$  to ion concentrations using coefficients.

### 9.6.3 Ion product and precipitation

$$Q_{sp} = [M^{m+}]^a [X^{x-}]^b$$

$Q_{sp} > K_{sp}$  means precipitation;  $Q_{sp} < K_{sp}$  means unsaturated;  $Q_{sp} = K_{sp}$  means saturated.

### 9.6.4 Common-ion solubility

$$K_{sp} = [M^{m+}][X^{x-}], \quad [X^{x-}]_{\text{initial}} \gg s \Rightarrow s \approx \frac{K_{sp}}{[X^{x-}]_{\text{initial}}}$$

A common ion lowers solubility by shifting dissolution toward the solid.

## 9.7 Exam recipe: $K_{sp}$ , molar solubility, common ion, hydroxide pH

**Use when:** the problem asks for solubility, precipitation, saturated solution pH, or common-ion effect.

**Steps:**

1. Write the dissolution reaction.
2. Write  $K_{sp}$  with coefficient powers.
3. Let  $s$  be molar solubility and express ion concentrations.
4. Include any initial common-ion concentration before adding  $s$ .
5. For hydroxides, convert  $[\text{OH}^-]$  to pOH and pH.

**Common traps:** forgetting the square/cube from coefficients; using  $s$  instead of  $2s$  for hydroxide concentration.

## 10 Electrochemistry and Oxidation States

### 10.1 Ctrl+F keywords: oxidation state, redox, galvanic cell, anode, cathode, Nernst, concentration cell, electrode potential, elektrokemi

Use for oxidation numbers, spontaneous galvanic cells, standard cell potentials, Nernst calculations, concentration cells, and electrochemical equilibrium.

### 10.2 Oxidation states

#### 10.2.1 Oxidation state rules

|                            |            |
|----------------------------|------------|
| free element               | 0          |
| monoatomic ion             | ion charge |
| oxygen in most compounds   | $-II$      |
| hydrogen in most compounds | $+I$       |
| halogens in binary salts   | $-I$       |
| sum in neutral molecule    | 0          |
| sum in polyatomic ion      | ion charge |

Oxidation states are bookkeeping numbers for redox. The element whose oxidation state increases is oxidized.

### 10.3 Exam recipe: find oxidation number

**Use when:** the problem asks for oxidation state of Cr, S, Mn, N, or another atom in a compound or ion.

**Steps:**

1. Assign fixed values first: group 1  $+I$ , group 2  $+II$ , oxygen usually  $-II$ , hydrogen usually  $+I$ .
2. Let the unknown oxidation state be  $x$ .
3. Set the sum equal to molecule charge or ion charge.
4. Solve for  $x$ .

**Fast checks:** free elements such as  $\text{O}_2$ ,  $\text{H}_2$ , metals have oxidation state 0.

### 10.4 Galvanic cells

#### 10.4.1 Anode, cathode, and electron flow

anode: oxidation, cathode: reduction,  $e^-$  : anode  $\rightarrow$  cathode

In a spontaneous galvanic cell, the cathode has the more positive reduction potential.

#### 10.4.2 Cell notation



Single bars are phase boundaries and the double bar is the salt bridge. Anode is written on the left by convention.

### 10.4.3 Standard cell potential

$$E_{\text{cell}}^{\circ} = E_{\text{cathode}}^{\circ} - E_{\text{anode}}^{\circ}$$

Use tabulated values as reduction potentials. Do not multiply  $E^{\circ}$  by stoichiometric coefficients.

### 10.4.4 Gibbs free energy and equilibrium

$$\Delta G^{\circ} = -nFE_{\text{cell}}^{\circ}, \quad nFE_{\text{cell}}^{\circ} = RT \ln K$$

Positive  $E_{\text{cell}}^{\circ}$  gives negative  $\Delta G^{\circ}$  and product-favored equilibrium.

## 10.5 Exam recipe: galvanic cell potential and spontaneous direction

**Use when:** the problem gives two half-cell reduction potentials or asks anode/cathode.

**Steps:**

1. Choose the more positive reduction potential as reduction at the cathode.
2. The other half-reaction is reversed for oxidation at the anode.
3. Compute  $E_{\text{cell}}^{\circ} = E_{\text{cathode}}^{\circ} - E_{\text{anode}}^{\circ}$ .
4. Balance electrons only for the reaction and  $n$ , not for scaling  $E^{\circ}$ .

**Common traps:** multiplying voltages by coefficients; swapping anode and cathode signs.

## 10.6 Nernst equation and concentration cells

### 10.6.1 Nernst equation

$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{RT}{nF} \ln Q$$

The reaction quotient is written for the balanced cell reaction.

### 10.6.2 Nernst equation at 298 K

$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.0592 \text{ V}}{n} \log Q$$

Use base-10 logarithm only with 0.0592 V. Use  $n$  as moles of electrons transferred.

### 10.6.3 Concentration cell

$$E_{\text{cell}}^{\circ} = 0, \quad E_{\text{cell}} = \frac{0.0592 \text{ V}}{n} \log \left( \frac{c_{\text{high}}}{c_{\text{low}}} \right)$$

The high-concentration half-cell is reduced and the low-concentration half-cell is oxidized for metal/metal-ion cells.

### 10.6.4 Solubility product from cell potential

$$\Delta G^{\circ} = -nFE_{\text{cell}}^{\circ} = -RT \ln K$$

Write the cell reaction so its equilibrium constant matches the desired dissolution or precipitation expression.

## 10.7 Exam recipe: Nernst concentration cell

**Use when:** identical electrodes have different ion concentrations, for example copper or nickel concentration cells.

**Steps:**

1. Set  $E_{\text{cell}}^{\circ} = 0$ .
2. Find  $n$  from the metal ion charge.
3. Put the larger ion concentration over the smaller one.
4. Compute  $E = (0.0592/n) \log(c_{\text{high}}/c_{\text{low}})$  at 298 K.

**Fast checks:** concentration-cell voltage is positive when written spontaneously and becomes zero when concentrations are equal.

## 11 Exam Strategy and Mixed Task Recipes

### 11.1 Ctrl+F keywords: multiple choice, statement validity, units, limiting case, answer check, traps, Danish keywords

Use for cross-topic exam tactics that recur in marked multiple-choice and calculation problems.

#### 11.2 Multiple-choice statement elimination

##### 11.2.1 One false clause makes an option false

valid answer option = all required clauses true

For statement options, test each clause independently: sign, unit, condition, and exception. Eliminate options with one wrong condition even if the formula name looks right.

##### 11.2.2 Dimensional and unit check

final unit must match requested quantity

Use units as algebraic factors. Common exam errors are Celsius in gas/Arrhenius equations, entropy in J with enthalpy in kJ, and cell length units in density.

##### 11.2.3 Logarithm checks for pH, $K$ , and Nernst

$$\text{pH} = -\log[\text{H}_3\text{O}^+], \quad \Delta G^\circ = -RT \ln K, \quad E = E^\circ - \frac{0.0592}{n} \log Q$$

A factor 10 in concentration changes pH or Nernst terms by one logarithmic unit. Check whether ln or log belongs with the numerical constant.

### 11.3 Recurring mixed recipes

#### 11.3.1 Ideal gas stoichiometry

$$n_{\text{gas}} = \frac{pV}{RT}$$

Convert gas data to moles before applying reaction coefficients. This links gas-law questions with limiting-reactant calculations.

#### 11.3.2 Percent change in hydronium concentration from pH

$$\frac{[\text{H}_3\text{O}^+]_2}{[\text{H}_3\text{O}^+]_1} = 10^{\text{pH}_1 - \text{pH}_2}, \quad \% \text{ change} = (\text{ratio} - 1)100\%$$

Use for ocean-acidification or pH-change wording. Lower pH means larger hydronium concentration.

#### 11.3.3 Recognition keywords, English and Danish

| exam wording                             | go to                   |
|------------------------------------------|-------------------------|
| bundfald, precipitate, sparingly soluble | $K_{sp}, Q_{sp}$        |
| puffer, buffer, Henderson                | acid/base buffer        |
| galvanisk, electrode, concentration cell | Nernst/electrochemistry |
| planar, bond angle, pyramidal            | Lewis/VSEPR             |
| limiting, excess, yield, udbytte         | stoichiometry           |
| vapor pressure, boiling point            | intermolecular forces   |
| rate law, half-life, Arrhenius           | kinetics                |

Use the wording to jump to the relevant recipe before doing algebra.